

Project Progress Report

(Group – 04)

Week 5: July 10 – July 16, 2025

1. Graph Classification

Graph Classification was successfully implemented using the MUTAG dataset. This task evaluates the model's ability to assign labels to entire graphs based on their structural and feature-based patterns.

2. Graph Regression

Graph Regression was implemented using a small synthetic graph dataset generated via PyTorch Geometric's built-in utilities. This task focuses on predicting continuous values for entire graphs.

3. Graph Matching

Graph Matching was implemented using a Siamese GCN model. The MUTAG dataset was used for experimentation, enabling the model to learn similarity measures between graph pairs.

4. Graph Reconstruction

A Graph Autoencoder (GAE) was implemented to perform Graph Reconstruction. The Cora dataset was used to train the model to learn compressed representations and reconstruct the original graph structure from these representations.

5. Graph Generation

Graph Generation was implemented using the Graph Variational Autoencoder (GraphVAE) model. The MUTAG dataset served as the sample input to train the model to generate new graphs with similar structural characteristics.